

Provable Diffusion Posterior Sampling for Bayesian Inversion

Chenguang Duan
RWTH Aachen

Joint work with **Jinyuan Chang** (SWUFE & CAS), **Yuling Jiao** (WHU), **Ruoxuan Li** (WHU),
Jerry Zhijian Yang (WHU), and **Cheng Yuan** (WHU)

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Outline

- ① Introduction to Inverse Problems
- ② Diffusion-based Posterior Sampling
- ③ Convergence Guarantee
- ④ Numerical Experiments
- ⑤ Concluding Remarks

Applications of Inverse Problems



(a) Medical Imaging



(b) Weather Prediction



(c) Image Processing

Mathematical Formulation of Bayesian Inverse Problems

Consider the measurement model:

$$\mathbf{Y} = \mathcal{F}(\mathbf{X}) + \mathbf{n}.$$

- ▶ $\mathcal{F} : \mathbb{R}^d \rightarrow \mathbb{R}^n$ is a **known** forward map.
- ▶ $\mathbf{X} \in \mathbb{R}^d$ is an **unknown** signal with prior density π .
- ▶ π is typically intractable but one has $\mathbf{X}_1, \dots, \mathbf{X}_N \sim^{\text{i.i.d.}} \pi$.
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Given noisy measurement $\mathbf{Y} = \mathbf{y}$, the goal of Bayesian inverse problem is to estimate the posterior density

$$q(\mathbf{x}|\mathbf{y}) \propto \exp(-\ell_{\mathbf{y}}(\mathbf{x}))\pi(\mathbf{x}),$$

where $\ell_{\mathbf{y}}(\cdot)$ is the negative log-likelihood.

Optimization-based approach

Maximum-a-posteriori (MAP): Find a signal that minimizes the posterior density:

$$\arg \max_{\mathbf{x}} \log q(\mathbf{x}|\mathbf{y}) = \arg \max_{\mathbf{x}} -\ell_{\mathbf{y}}(\mathbf{x}) + \log \pi(\mathbf{x})$$

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- ▶ Point estimation, cannot quantify the uncertainty

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Sampling from the posterior density $q(\cdot|\mathbf{y})$:

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- ▶ Uncertainty quantification

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- ▶ Manual selection of the prior: cannot capture the complex structure of the prior distribution.
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Prior Score + Diffusion Models represent a highly promising solution!

Diffusion Posterior Sampling

Forward process

$$d\mathbf{X}_t = -\mathbf{X}_t dt + \sqrt{2}d\mathbf{B}_t, \quad \mathbf{X}_0 \sim \pi, \quad t \in (0, T).$$

Time-reversal process

$$d\bar{\mathbf{X}}_t = (\bar{\mathbf{X}}_t + 2\nabla_{\mathbf{x}} \log \pi_{T-t}(\bar{\mathbf{X}}_t))dt + \sqrt{2}d\mathbf{B}_t, \quad \bar{\mathbf{X}}_0 \sim \pi_T, \quad t \in (0, T). \quad \bar{\mathbf{X}}_t \sim \pi_{T-t}$$

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Posterior time-reversal process

$$d\bar{\mathbf{X}}_t^y = (\bar{\mathbf{X}}_t^y + 2\nabla_{\mathbf{x}} \log q_{T-t}(\bar{\mathbf{X}}_t^y | \mathbf{y}))dt + \sqrt{2}d\mathbf{B}_t, \quad \bar{\mathbf{X}}_0^y \sim q_T(\cdot | \mathbf{y}) \quad \bar{\mathbf{X}}_t \sim q_{T-t}(\cdot | \mathbf{y})$$

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How to estimate the posterior score $\nabla \log q_t(\cdot | \mathbf{y})$ for $t \in (0, T)$?

Existing Approximation Methods for Posterior Score Estimation

$$\nabla \log q_t(\mathbf{x}_t|\mathbf{y}) = \nabla \log p(\mathbf{y}|\mathbf{x}_t) + \nabla \log \pi_t(\mathbf{x}_t)$$

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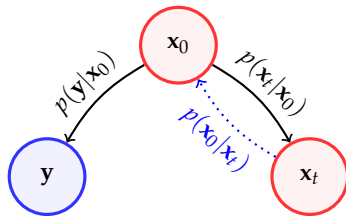
$$\begin{aligned} p(\mathbf{y}|\mathbf{x}_t) &= \int p(\mathbf{y}|\mathbf{x}_0, \mathbf{x}_t) p(\mathbf{x}_0|\mathbf{x}_t) d\mathbf{x}_0 \\ &= \int \exp(-\ell_{\mathbf{y}}(\mathbf{x}_0)) p(\mathbf{x}_0|\mathbf{x}_t) d\mathbf{x}_0 \end{aligned}$$

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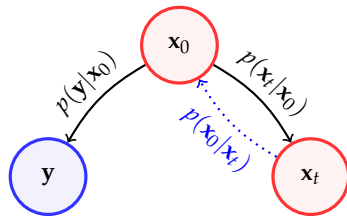
The time-dependent likelihood $p(\mathbf{y}|\mathbf{x}_t)$ is intractable.



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Approximation Methods

DPS (Hyungjin Chung et al. 2023)

LGD (Jiaming Song et al. 2023)

...

Dirac approximation

Gaussian approximation

...

$$p(\mathbf{x}_0|\mathbf{x}_t) \approx \delta_{\mathbb{E}[\mathbf{X}_0|\mathbf{x}_t]}(\mathbf{x}_0)$$

$$p(\mathbf{x}_0|\mathbf{x}_t) \approx N(\mathbb{E}[\mathbf{X}_0|\mathbf{x}_t], r_t^2 \mathbf{I}_d)$$

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Posterior Score Estimation via Monte Carlo

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Conditional Tweedie's formula

For each $t \in (0, T)$, the score function of the posterior density satisfies

$$\nabla_{\mathbf{x}} \log q_t(\mathbf{x} | \mathbf{y}) = -\frac{1}{\sigma_t^2} \mathbf{x} + \frac{\mu_t}{\sigma_t^2} \mathbf{D}(t, \mathbf{x}, \mathbf{y}), \quad (\mathbf{x}, \mathbf{y}) \in \mathbb{R}^d \times \mathbb{R}^n,$$

where the posterior denoiser $\mathbf{D}(t, \mathbf{x}, \mathbf{y})$ is defined as

$$\mathbf{D}(t, \mathbf{x}, \mathbf{y}) := \mathbb{E}[\mathbf{X}_0 | \mathbf{X}_t = \mathbf{x}, \mathbf{Y} = \mathbf{y}].$$

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Expectation with respect to

$$p_t(\mathbf{x}_0 | \mathbf{x}, \mathbf{y}) \propto \pi(\mathbf{x}_0) \exp \left(-\frac{\|\mathbf{x} - \mu_t \mathbf{x}_0\|_2^2}{2\sigma_t^2} - \ell_{\mathbf{y}}(\mathbf{x}_0) \right)$$

Sampling from the posterior denoising density

- Langevin dynamics

$$d\mathbf{X}_{0,s}^{\mathbf{x},\mathbf{y},t} = \left(\nabla \log \pi(\mathbf{X}_{0,s}^{\mathbf{x},\mathbf{y},t}) + \frac{\mu_t}{\sigma_t^2} (\mathbf{x} - \mu_t \mathbf{X}_{0,s}^{\mathbf{x},\mathbf{y},t}) - \nabla \ell_{\mathbf{y}}(\mathbf{X}_{0,s}^{\mathbf{x},\mathbf{y},t}) \right) ds + \sqrt{2} d\mathbf{B}_s, \quad s \in (0, S)$$

- Langevin dynamics with **estimated prior score**

$$d\hat{\mathbf{X}}_{0,s}^{\mathbf{x},\mathbf{y},t} = \left(\hat{\mathbf{s}}_{\pi}(\hat{\mathbf{X}}_{0,s}^{\mathbf{x},\mathbf{y},t}) + \frac{\mu_t}{\sigma_t^2} (\mathbf{x} - \mu_t \hat{\mathbf{X}}_{0,s}^{\mathbf{x},\mathbf{y},t}) - \nabla \ell_{\mathbf{y}}(\hat{\mathbf{X}}_{0,s}^{\mathbf{x},\mathbf{y},t}) \right) ds + \sqrt{2} d\mathbf{B}_s, \quad s \in (0, S).$$

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Monte Carlo approximation

- Estimation of the posterior score

$$\hat{\mathbf{s}}_m^S(t, \mathbf{x}, \mathbf{y}) \triangleq -\frac{\mathbf{x}}{\sigma_t^2} + \frac{\mu_t}{\sigma_t^2} \frac{1}{m} \sum_{i=1}^m \hat{\mathbf{X}}_{0,S,i}^{\mathbf{x},\mathbf{y},t} \approx \nabla_{\mathbf{x}} \log q_t(\mathbf{x}|\mathbf{y}), \quad \hat{\mathbf{X}}_{0,S,1}^{\mathbf{x},\mathbf{y},t}, \dots, \hat{\mathbf{X}}_{0,S,m}^{\mathbf{x},\mathbf{y},t} \sim^{\text{i.i.d.}} \hat{p}_t^S(\cdot|\mathbf{x}, \mathbf{y}).$$

Algorithm 1: Posterior score estimation via Monte Carlo

Input: Measurement $\mathbf{y}$, prior score estimator $\hat{\mathbf{s}}_\pi$, noise level t , evaluation point $\mathbf{x}$, simulation horizon S , initial density $\hat{p}_t^0(\cdot|\mathbf{y})$ for Langevin dynamics.

Output: Posterior score estimator $\hat{\mathbf{s}}_m^S(t, \mathbf{x}, \mathbf{y})$.

- 1 **for** $i = 1, \dots, m$ **do**
 - 2 Initialize particle by $\hat{\mathbf{X}}_{0,0,i}^{\mathbf{x},\mathbf{y},t} \sim \hat{p}_t^0(\cdot|\mathbf{x}, \mathbf{y})$.
 - 3 Evolve particle to time S by simulating Langevin dynamics to obtain $\hat{\mathbf{X}}_{0,S,i}^{\mathbf{x},\mathbf{y},t}$.
 - 4 **end**
 - 5 Compute the posterior denoiser estimate $\hat{\mathbf{D}}_m^S(t, \mathbf{x}, \mathbf{y})$ using $\{\hat{\mathbf{X}}_{0,S,i}^{\mathbf{x},\mathbf{y},t}\}_{i=1}^m$ via Monte Carlo method.
 - 6 Compute the posterior score estimate $\hat{\mathbf{s}}_m^S(t, \mathbf{x}, \mathbf{y})$ using $\hat{\mathbf{D}}_m^S(t, \mathbf{x}, \mathbf{y})$.
 - 7 **return** $\hat{\mathbf{s}}_m^S(t, \mathbf{x}, \mathbf{y})$
-

► **A1.** Semi-log-concavity of the posterior distribution.

For a fixed $\mathbf{y} \in \mathbb{R}^n$, there exist a constant $\alpha > 0$, such that $q_0(\cdot|\mathbf{y}) \in C^2(\mathbb{R}^d)$ satisfies $-\nabla^2 \log q_0(\mathbf{x}_0|\mathbf{y}) \succeq -\alpha \mathbf{I}_d$ for each $\mathbf{x}_0 \in \mathbb{R}^d$.

Log-concavity of posterior denoising density

Suppose **A1** holds, then for each $t > 0$,

$$\left(\frac{\mu_t^2}{\sigma_t^2} - \alpha\right) \mathbf{I}_d \preceq -\nabla_{\mathbf{x}_0}^2 \log p_t(\mathbf{x}_0|\mathbf{x}, \mathbf{y}), \quad (\mathbf{x}_0, \mathbf{x}) \in \mathbb{R}^d \times \mathbb{R}^d.$$

Further, the posterior denoising density is log-concave provided that $t < \bar{t} := \frac{1}{2} \log(1 + \alpha^{-1})$.

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Monte Carlo-based approach for posterior score estimation works for small $t > 0$.

Warm-Start Strategy

Goal of warm-start: Sample from the terminal posterior density $q_T(\cdot|\mathbf{y})$.

$$d\bar{\mathbf{X}}_t^{\mathbf{y}} = (\bar{\mathbf{X}}_t^{\mathbf{y}} + 2\nabla_{\mathbf{x}} \log q_{T-t}(\bar{\mathbf{X}}_t^{\mathbf{y}}|\mathbf{y}))dt + \sqrt{2}d\mathbf{B}_t, \quad \underbrace{\bar{\mathbf{X}}_0^{\mathbf{y}} \sim q_T(\cdot|\mathbf{y})}_{\text{warm-start}}$$

- ▶ For **sufficiently large** T , we can use Gaussian approximation $q_T(\cdot|\mathbf{y}) \approx N(\mathbf{0}, \mathbf{I}_d)$.
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Warm-start strategy

Langevin dynamics with estimated posterior score:

$$d\hat{\mathbf{X}}_{T,u}^{\mathbf{y}} = \hat{\mathbf{s}}_m^S(T, \hat{\mathbf{X}}_{T,u}^{\mathbf{y}}, \mathbf{y})du + \sqrt{2}d\mathbf{B}_u, \quad u \in (0, U).$$

Here $\hat{\mathbf{s}}_m^S(T, \cdot, \mathbf{y})$ is obtained by Monte Carlo-based posterior score estimation.

Algorithm 2: Warm-start for diffusion posterior sampling

Input: Measurement $\mathbf{y}$, simulation horizon U , terminal density $\hat{q}_T^0(\cdot|\mathbf{y})$ for Langevin dynamics.

Output: Sample $\hat{\mathbf{X}}_{T,U}^{\mathbf{y}}$ approximately distributed according to $q_T(\cdot|\mathbf{y})$.

- 1 Initialize particle by $\hat{\mathbf{X}}_{T,0}^{\mathbf{y}} \sim \hat{q}_T^0(\cdot|\mathbf{y})$.
 - 2 Simulate the approximate Langevin dynamics from 0 to U using the estimated posterior score function $\hat{\mathbf{s}}_m^S(t, \mathbf{x}, \mathbf{y})$ to obtain $\hat{\mathbf{X}}_{T,U}^{\mathbf{y}}$.
 - 3 **return** $\hat{\mathbf{X}}_{T,U}^{\mathbf{y}}$
-

► **A2.** Sub-Gaussian tails.

The log-prior density $\log \pi_0 \in C^2(\mathbb{R}^d)$, and there exist $V_{\text{SG}} > 0$ and $C_{\text{SG}} > 0$, such that

$$\int \exp\left(\frac{\|\mathbf{x}_0\|_2^2}{V_{\text{SG}}^2}\right) \pi_0(\mathbf{x}_0) d\mathbf{x}_0 \leq C_{\text{SG}}.$$

Log-Sobolev inequality of posterior

Suppose **A2** holds, then for each $T > \underline{t} := \frac{1}{2} \log(1 + 2V_{\text{SG}}^2)$,

$$C_{\text{LSI}}(q_T(\cdot|\mathbf{y})) \leq 12\sigma_T^2 \exp\left(2\frac{\sigma_T^2 + 2\mu_T^2 V_{\text{SG}}^2}{\sigma_T^2 - 2\mu_T^2 V_{\text{SG}}^2} \log(\kappa_{\mathbf{y}}^2 C_{\text{SG}}^2)\right).$$

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The terminal posterior density $q_T(\cdot|\mathbf{y})$ satisfies LSI for large T .

Diffusion Posterior Sampling

$$d\hat{\mathbf{X}}_t^y = (\hat{\mathbf{X}}_t^y + 2 \underbrace{\hat{\mathbf{s}}_m^S(T-t, \hat{\mathbf{X}}_t^y, \mathbf{y})}_{\text{posterior score estimation}})dt + \sqrt{2}d\mathbf{B}_t, \quad \underbrace{\hat{\mathbf{X}}_0^y \sim \hat{q}_T^U(\cdot|\mathbf{y})}_{\text{warm-start}}, \quad t \in (0, T - T_0),$$

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Algorithm 4: Diffusion-based posterior sampling

Input: Measurement $\mathbf{y}$, diffusion terminal time T , early-stopping time T_0 .

Output: Posterior sample $\hat{\mathbf{X}}_{T-T_0}^y$ approximately drawn from $q_0(\cdot|\mathbf{y})$.

- 1 **Warm-start:** Set the initial particle $\hat{\mathbf{X}}_0^y = \hat{\mathbf{X}}_{T,U}^y$, where $\hat{\mathbf{X}}_{T,U}^y$ is a warm-start sample.
 - 2 **Reverse diffusion:** Simulate the approximate time-reversal stochastic process over the time interval $[0, T - T_0]$, employing the estimated score function $\hat{\mathbf{s}}_m^S(t, \mathbf{x}, \mathbf{y})$, to generate $\hat{\mathbf{X}}_{T-T_0}^y$.
 - 3 **return** $\hat{\mathbf{X}}_{T-T_0}^y$
-

Diffusion Posterior Sampling

$$d\hat{\mathbf{X}}_t^y = (\hat{\mathbf{X}}_t^y + 2 \underbrace{\hat{\mathbf{s}}_m^S(T-t, \hat{\mathbf{X}}_t^y, \mathbf{y})}_{\text{posterior score estimation}})dt + \sqrt{2}d\mathbf{B}_t, \quad \underbrace{\hat{\mathbf{X}}_0^y \sim \hat{q}_T^U(\cdot|\mathbf{y})}_{\text{warm-start}}, \quad t \in (0, T - T_0),$$

- Posterior score estimation: $T \in (0, \bar{t})$
- Warm-start: $T \in (\underline{t}, +\infty)$

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*Does there exist a terminal time T such that both the **posterior score estimation** and the **terminal posterior sampling procedures** are theoretically guaranteed to converge?*

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Does there exist a terminal time T such that both the *posterior score estimation* and the *terminal posterior sampling procedures* are theoretically guaranteed to converge?

Duality of Log-Concavity

Suppose **A1** and **A2** hold. If $2\alpha V_{\text{SG}}^2 \leq 1$, then there exists a terminal time T such that:

- the posterior denoising density $p_t(\cdot|\mathbf{x}, \mathbf{y})$ is log-concave for all $t \in (0, T)$; and
- the posterior density $q_T(\cdot|\mathbf{y})$ satisfies LSI.

Outline

- ① Introduction to Inverse Problems
- ② Diffusion-based Posterior Sampling
- ③ Convergence Guarantee**
- ④ Numerical Experiments
- ⑤ Concluding Remarks

Error Decomposition

The posterior time-reversal process

$$d\bar{\mathbf{X}}_t^{\mathbf{y}} = (\bar{\mathbf{X}}_t^{\mathbf{y}} + 2\nabla_{\mathbf{x}} \log q_{T-t}(\bar{\mathbf{X}}_t^{\mathbf{y}}|\mathbf{y}))dt + \sqrt{2}d\mathbf{B}_t, \quad \bar{\mathbf{X}}_0^{\mathbf{y}} \sim q_T(\cdot|\mathbf{y}), \quad t \in (0, T).$$

The approximate time-reversal process

$$d\hat{\mathbf{X}}_t^{\mathbf{y}} = (\hat{\mathbf{X}}_t^{\mathbf{y}} + 2 \underbrace{\hat{\mathbf{s}}_m^S(T-t, \hat{\mathbf{X}}_t^{\mathbf{y}}, \mathbf{y})}_{\text{posterior score estimation}})dt + \sqrt{2}d\mathbf{B}_t, \quad \underbrace{\hat{\mathbf{X}}_0^{\mathbf{y}} \sim \hat{q}_T^U(\cdot|\mathbf{y})}_{\text{warm-start}}, \quad \underbrace{t \in (0, T - T_0)}_{\text{early-stopping}}.$$

Sources of errors

- ▶ posterior score estimation error
- ▶ warm-start error
- ▶ early-stopping error

Additional Assumptions

- ▶ **A3.** For a fixed measurement $\mathbf{y} \in \mathbb{R}^n$, there exist constants $r \geq 1$, $B > 1$ and $L > 1$ such that for each $\mathbf{x}_0 \in \mathbb{R}^d$,

$$\|\nabla \log \pi_0(\mathbf{x}_0)\|_2 \leq B(1 + \|\mathbf{x}_0\|_2^r).$$

- ▶ **A4.** For a fixed measurement $\mathbf{y} \in \mathbb{R}^n$, there exists a constant $H > 1$, such that $\|\nabla \log q_0(\mathbf{0}|\mathbf{y})\|_2 \leq H$.

- ▶ **A5.** Let $\hat{\mathbf{s}}_{\text{prior}}$ be an estimator of the prior score function $\nabla \log \pi_0$. There exists $\varepsilon_{\text{prior}} \in (0, 1)$, such that

$$\int \|\nabla \log \pi_0(\mathbf{x}_0) - \hat{\mathbf{s}}_{\text{prior}}(\mathbf{x}_0)\|_2^2 \pi_0(\mathbf{x}_0) d\mathbf{x}_0 \leq \varepsilon_{\text{prior}}^2.$$

- ▶ **A6.** For a fixed measurement $\mathbf{y} \in \mathbb{R}^n$, there exist constants $G > 1$, such that $\hat{\mathbf{s}}_{\text{prior}} - \nabla \ell_{\mathbf{y}}$ is G -Lipschitz.

Error Decomposition

Suppose **A2** hold.

- Let $T_0 \in (0, \frac{1}{2})$ and $R^2 = (4\mu_{T_0}^2 V_{SG}^2 + 16\sigma_{T_0}^2) \log(\kappa_y \varepsilon_{T_0}^{-1})$. For $\varepsilon_{T_0} \in (0, 1)$, we have

$$\mathbb{W}_2^2(q_0(\cdot|\mathbf{y}), \mathcal{M}(\mu_{T_0}^{-1}) \# \hat{q}_{T-T_0}^R(\cdot|\mathbf{y})) \leq C \left\{ \frac{\sigma_{T_0}^2}{\mu_{T_0}^2} + \varepsilon_{T_0} \log \left(\frac{\kappa_y}{\varepsilon_{T_0}} \right) \right\},$$

provided that $\|q_{T_0}(\cdot|\mathbf{y}) - \hat{q}_{T-T_0}(\cdot|\mathbf{y})\|_{TV}^2 = \varepsilon_{T_0}^2$. Here C is a constant only depending on d , V_{SG} , and C_{SG} .

- For $0 < T_0 < T$, it holds that

$$\begin{aligned} & \|q_{T_0}(\cdot|\mathbf{y}) - \hat{q}_{T-T_0}(\cdot|\mathbf{y})\|_{TV}^2 \\ & \leq \underbrace{\int_{T_0}^T \mathbb{E} [\|\nabla \log q_t(\mathbf{X}_t^y|\mathbf{y}) - \hat{\mathbf{s}}_m^S(t, \mathbf{X}_t^y, \mathbf{y})\|_2^2] dt}_{\text{posterior score estimation}} + \underbrace{2\|q_T(\cdot|\mathbf{y}) - \hat{q}_0^U(\cdot|\mathbf{y})\|_{TV}^2}_{\text{warm-start}}. \end{aligned}$$

Error of Posterior Score Estimation

Suppose **A1**, **A2**, **A3**, **A5**, and **A6** hold. Let $0 < T_0 < T < \frac{1}{2} \log(1 + \alpha^{-1})$. For each time $t \in (T_0, T)$,

$$\begin{aligned}
 & \mathbb{E} [\|\nabla \log q_t(\mathbf{X}_t^y | \mathbf{y}) - \hat{\mathbf{s}}_m^S(t, \mathbf{X}_t^y, \mathbf{y})\|_2^2] \\
 & \leq \underbrace{C \frac{\mu_{T_0}^2}{\sigma_{T_0}^4} \frac{\kappa_y}{m}}_{\text{Monte Carlo}} + \underbrace{C \frac{\mu_{T_0}^2}{\sigma_{T_0}^4} \exp\left(-\frac{2(\mu_T^2 - \alpha\sigma_T^2)}{\sigma_T^2} S\right) \eta_y^2}_{\text{convergence of Langevin dynamics}} \\
 & \quad + \underbrace{C \frac{\mu_{T_0}^2}{\sigma_{T_0}^4} \left(\frac{\sigma_T^2 \eta_y^2}{\mu_T^2 - \alpha\sigma_T^2} + S\right)^{\frac{1}{2}} \exp\left(2\left(G + \frac{\mu_{T_0}^2}{\sigma_{T_0}^2}\right) S\right) \kappa_y^{\frac{1}{2}} \varepsilon_{\text{prior}}^{\frac{1}{2}}}_{\text{prior score estimation error}},
 \end{aligned}$$

where C is a constant only depending on d , B , V_{SG} , and C_{SG} . Here κ_y is the condition number, and η_y is the initial discrepancy of Langevin dynamics.

Suppose **A1**, **A2**, **A3**, **A4**, **A5**, and **A6** hold. Assume $2\alpha V_{SG}^2 \leq 1$. Let the terminal time satisfy $T \in (\frac{1}{2} \log(1 + 2V_{SG}^2), \frac{1}{2} \log(1 + \alpha^{-1}))$. Then for $\varepsilon \in (0, 1)$, it holds that

$$\|q_T(\cdot|\mathbf{y}) - \hat{q}_T^U(\cdot|\mathbf{y})\|_{TV}^2 \leq C\varepsilon,$$

provided that

$$U = \Theta\left(\exp\left(2\frac{\sigma_T^2 + 2\mu_T^2 V_{SG}^2}{\sigma_T^2 - 2\mu_T^2 V_{SG}^2} \log(\kappa_{\mathbf{y}}^2 C_{SG}^2)\right) \log\left(\frac{\zeta_{\mathbf{y}}^2}{\varepsilon}\right)\right),$$

$$\varepsilon_{\text{post}} = \Theta\left(\exp\left(-2\frac{\sigma_T^2 + 2\mu_T^2 V_{SG}^2}{\sigma_T^2 - 2\mu_T^2 V_{SG}^2} \log(\kappa_{\mathbf{y}}^2 C_{SG}^2)\right) \left(\zeta_{\mathbf{y}}^2 + \log\left(\frac{\zeta_{\mathbf{y}}^2}{\varepsilon}\right)\right)^{-1} (\mu_T^2 - \alpha\sigma_T^2)^3 \frac{\varepsilon^2}{\kappa_{\mathbf{y}}^{1/2}}\right),$$

where C is a constant only depending on d , B , H , V_{SG} , and C_{SG} . The initial divergence $\zeta_{\mathbf{y}}^2$ and the posterior score matching error at the terminal time T are defined, respectively, as:

$$\zeta_{\mathbf{y}}^2 := \chi^2(\hat{q}_T^0(\cdot|\mathbf{y})\|q_T(\cdot|\mathbf{y})), \quad \varepsilon_{\text{post}}^2 := \mathbb{E}[\|\nabla \log q_T(\mathbf{X}_T^{\mathbf{y}}|\mathbf{y}) - \hat{\mathbf{s}}_m^S(T, \mathbf{X}_T^{\mathbf{y}}, \mathbf{y})\|_2^2].$$

Convergence of Posterior Sampling

Suppose **A1**, **A2**, **A3**, **A4**, **A5**, and **A6** hold. Assume $2\alpha V_{\text{SG}}^2 \leq 1$. Let the terminal time satisfy $T \in (\frac{1}{2} \log(1 + 2V_{\text{SG}}^2), \frac{1}{2} \log(1 + \alpha^{-1}))$. For each time $t \in (T_0, T)$, Then for $\varepsilon \in (0, 1)$, the following inequality holds:

$$\begin{aligned} & \mathbb{E} [\mathbb{W}_2^2(q_0(\cdot|\mathbf{y}), \mathcal{M}(\mu_{T_0}^{-1}) \# \hat{q}_{T-T_0}^R(\cdot|\mathbf{y})))] \\ & \leq C \left\{ \frac{\sigma_{T_0}^2}{\mu_{T_0}^2} + \left(\frac{\mu_{T_0}}{\sigma_{T_0}^2} \varepsilon + \varepsilon^{\frac{1}{2}} \right) \log \left(\kappa_{\mathbf{y}} \left(\frac{\mu_{T_0}}{\sigma_{T_0}^2} \varepsilon + \varepsilon^{\frac{1}{2}} \right)^{-1} \right) \right\}, \end{aligned}$$

where C is a constant only depending on d , $\kappa_{\mathbf{y}}$, B , H , V_{SG} , and C_{SG} . Furthermore, by setting $T_0 = \sqrt{\varepsilon_0}$, we have

$$\mathbb{E} [\mathbb{W}_2^2(q_0(\cdot|\mathbf{y}), \mathcal{M}(\mu_{T_0}^{-1}) \# \hat{q}_{T-T_0}^R(\cdot|\mathbf{y})))] \leq C' \varepsilon^{\frac{1}{2}} \log \varepsilon^{-1}.$$

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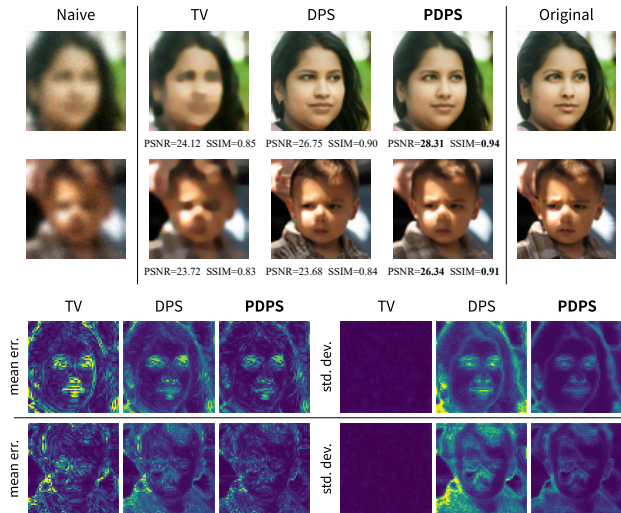
Numerical Experiments

Dataset: FFHQ64

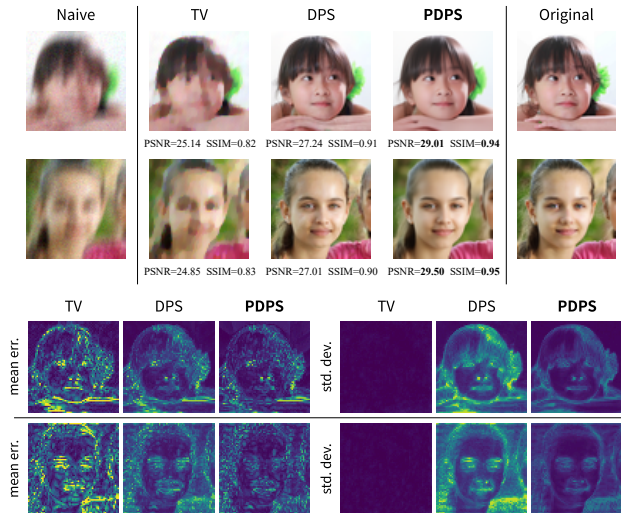


	Gaussian Deblur			Motion Deblur			Nonlinear Deblur		
	TV	DPS	Ours	TV	DPS	Ours	TV	DPS	Ours
PSNR	23.95	24.15	26.42	24.65	26.66	28.86	19.70	20.93	28.44
SSIM	0.81	0.81	0.87	0.80	0.88	0.92	0.53	0.68	0.91

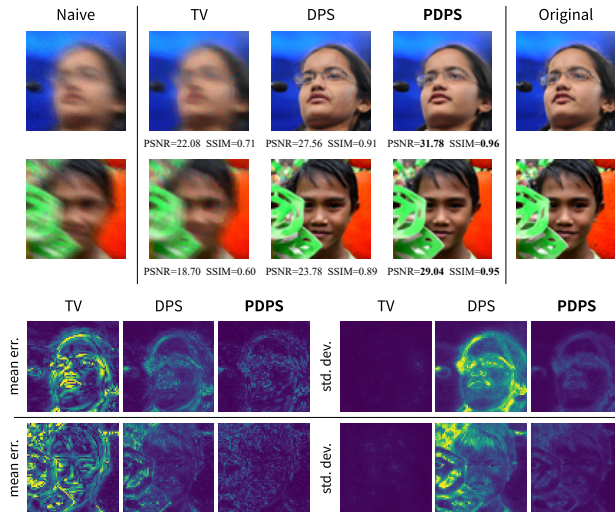
Gaussian Deblurring



Motion Deblurring

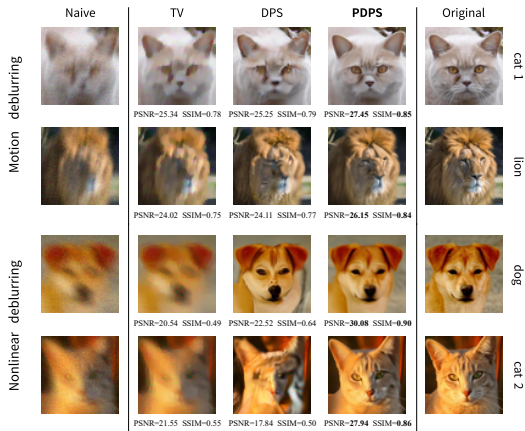


Nonlinear Deblurring

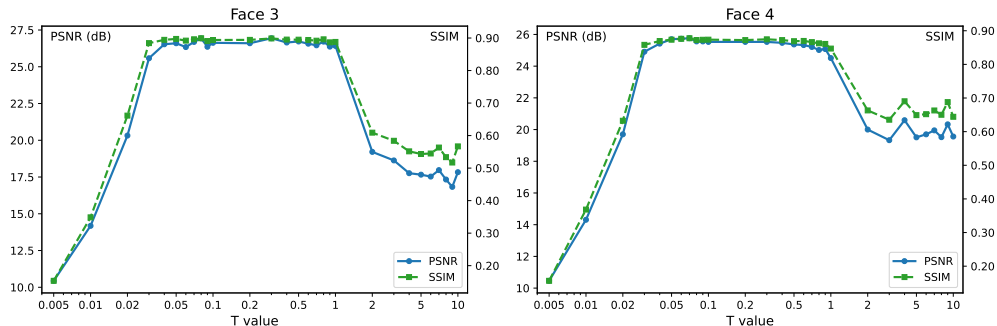


Cross-Dataset Generalization Assessment

- Prior score is learning using FFHQ64.
- Groundtruth image is from AFHQ64.



Ablation Study of Terminal Time



- For too small T , the terminal posterior density $q_T(\cdot|\mathbf{y})$ does not satisfies LSI.
- For too large T , the posterior denoising density $p_t(\cdot|\mathbf{x}, \mathbf{y})$ is non-log-concave for t close to T .

Concluding Remarks

Summary

- ▶ A diffusion-based posterior sampling method without heuristic approximations.
- ▶ Non-asymptotic error analysis for posterior sampling.
- ▶ Experimental performance in image restoration.

Concluding Remarks

Summary

- ▶ A diffusion-based posterior sampling method without heuristic approximations.
- ▶ Non-asymptotic error analysis for posterior sampling.
- ▶ Experimental performance in image restoration.

Further work

- ▶ Derivative-free Bayesian inference.
- ▶ Blind inverse problems.
- ▶ Inverse problem in infinite-dimensional problems.

Reference

Jinyuan Chang, Chenguang Duan, Yuling Jiao, Ruoxuan Li, Jerry Zhijian Yang, and Cheng Yuan.
Provable Diffusion Posterior Sampling for Bayesian Inversion. arXiv:2512.08022

Thanks for your attention!

Email: duan@igpm.rwth-aachen.de

Homepage:

<https://chenguangduan.github.io/>

Google Scholar:

<https://scholar.google.com/citations?user=RpmGgyMAAAAJ>

